

Contour

A drawn-envelope filter: four hand-drawn curves move cutoff, resonance, volume and pan in time with the music.



Overview

Contour is a filter driven by curves you draw. Four curves run at once - CUTOFF, RESONANCE, VOLUME and PAN - each looping over a length you set, from a 1/32 note to 32 bars. Draw a pluck, a sweep, a stutter or a rhythmic gate, and Contour plays it back in time with the project, or restarts it on every MIDI note.

The filter itself is a 4x oversampled state-variable design with low-pass, high-pass, band-pass and notch modes. A diode-style soft clipper sits inside its feedback loop, so DRIVE adds analogue-style grit and resonance stays musical when pushed.

Each curve has its own amount knob, so you can switch any of them on or off without losing the drawing, and DEPTH scales all four together.

Reach for it for

- Rhythmic filter sweeps and plucks on pads, synths and loops

- Tempo-synced volume gating and ducking shapes (draw the dips)
- Auto-panning with any shape you like, not just a sine
- Triggered filter envelopes from MIDI notes (MIDI mode, one-shot)
- Wide, moving stereo filtering with STEREO offset

Controls

Curve tabs and the editor

The four tabs - CUTOFF, RESONANCE, VOLUME and PAN - pick which curve you are editing. The light on each tab shows that curve is in use (its amount knob is not at 0). The curve being edited is drawn filled in its colour; the other three show faintly behind it. The grid divides one cycle into 16 steps, with every fourth line brighter. The white line is the playhead; a fainter second line shows the right channel's place when STEREO is up. Top right, the current RATE, trigger mode and LOOP / ONE-SHOT are shown.

Click	Adds a point (up to 32 per curve). Keep the mouse down to drag it straight into place.
Drag a point	Moves it. The first and last points stay at the edges of the cycle; you can only move them up and down.
Double-click a point	Removes it (not the first or last).
Drag a line	Bends the line between two points into a curve. Dragging up lifts the middle of the line, whichever way it slopes.
Double-click a line	Straightens it again.
Right-click	A menu: Reset this curve (back to its starting shape), Flat (top), Flat (middle), Flat (bottom), Copy curve and Paste curve. Use copy and paste to move a shape from one tab to another.

The starting shapes: CUTOFF is a pluck (falls quickly from the top to the bottom), RESONANCE and VOLUME are flat, and PAN moves from centre to the right, across to the left and back to centre. The curves are saved with the project.

Filter

MODE	LP 6, LP 12, LP 18, LP 24 (low-pass, 6 to 24 dB per octave), HP 12 (high-pass), BP 12 (band-pass) or NOTCH. Starts on LP 12.
CUTOFF	The filter's base frequency, 20 Hz to 20 kHz. The cutoff curve moves the filter up (or down) from here.
RESO	Resonance, 0 - 100%. Higher settings add a whistling peak at the cutoff.
DRIVE	Pushes the input harder into the filter, 0 to +24 dB, so the soft clipper inside the filter adds grit. The level is compensated, so it changes the character more than the loudness.

Curve amounts

Each amount knob sits under a stripe in its curve's colour. At 0 that curve does nothing.

CUTOFF	How far the cutoff curve moves the filter, -8 to +8 octaves (starts at +4). The bottom of the curve is the CUTOFF knob's setting and the top is CUTOFF plus this amount. Negative amounts turn the curve upside down.
RESO	How far the resonance curve moves the resonance, -100 to +100%, added to the RESO knob.
VOLUME	How far the volume curve can turn the level down, 0 - 100%. The top of the curve is full level; the bottom is the level minus this amount. Draw dips to duck or gate.
PAN	How far the pan curve moves the sound, 0 - 100%. The middle of the curve is centre; the top pans to the right and the bottom to the left, fully at 100%.
DEPTH	Scales all four amounts at once, 0 - 100%, without changing the drawings. Good for automating the whole effect in and out.
STEREO	Moves the right channel's place in the curves later, up to half a cycle (0 - 50%). The cutoff, resonance and volume

then move differently on each side, for wide, moving filtering. (PAN follows the left channel's place.)

Timing

RATE	The length of one cycle of the curves: 1/32, 1/16, 1/8, 1/4, 1/2, 1 BAR, 2, 4, 8, 16 or 32 BARS. Starts at 1 BAR.
SYNC	The curves follow the project's timeline, so they line up with the bars wherever you start playback. When the project is stopped, they keep running at the project tempo.
MIDI	Every MIDI note-on restarts the curves from the beginning. Put Contour on a track that receives MIDI (or send MIDI to it) and each note triggers the shape, like a filter envelope. MIDI passes through unchanged.
FREE	The curves run at the project tempo, without following the timeline - even when the project is stopped.
LOOP / ONE-SHOT	LOOP repeats the curves. ONE-SHOT (with MIDI) plays them once per note and then holds the end of the curve until the next note.
TRIGGER	Restarts the curves by hand. In SYNC mode while playing, it moves where the cycle starts to the current position.
MIDI NOTE IN	Lights when a MIDI note arrives.

IN / OUT

TRIM	Input level before anything else, from -24 to +12 dB.
GUARD	Click the TRIM knob (without moving it) to switch the guard; its light sits at the knob's upper right. Off by default on Contour. When on, it gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the filter. It never boosts quiet sources. The light glows brighter while it is catching peaks.

MIX	Dry / filtered balance, 0 - 100%. The VOLUME and PAN curves act on the mixed signal.
OUTPUT	Output level, -24 to +12 dB.
SAT	Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
OFF / CLIP / LIMIT	The final stage: nothing, a soft clipper (no latency), or a lookahead limiter that holds the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER). CLIP is on by default; its key flashes red while it is clipping.
IN / OUT meters	Input and output level.

The header

S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Using MIDI triggering

1. Put Contour on the audio track you want to shape (for example a pad or a synth).
2. Give that track MIDI: draw a MIDI item on it, or send MIDI to it from another track (in the send, set the MIDI to go to the track).
3. Choose MIDI in the TIMING section, and ONE-SHOT if you want the shape to play once per note.
4. Play: MIDI NOTE IN lights and the curves restart on every note-on. Set RATE to the length you want the shape to take.

Tips

- For a classic synth pluck, use the default CUTOFF curve, MODE LP 24, a low CUTOFF and RATE 1/8 or 1/16.
- For a sidechain-style pump without a key input, draw a dip at the start of the VOLUME curve, set VOLUME to 60 - 100% and RATE to 1/4.
- Bend the lines rather than adding lots of points: a single bent line gives a smooth exponential sweep.
- Draw the shape once, then use DEPTH to bring it in gradually through a song.
- Small STEREO settings (5 - 15%) widen the filter movement without making it sound like two separate effects.

File: gfx-contour.jsfx